

EE 313: Measurement and Instrumentation

Course Code: EE 313

Course Name: Measurement and Instrumentation

L-T-P-C: 2-1-0-3

Prerequisites: IC 160: Basic Electrical Engineering and IC 161 Basic Electronics

Engineering Intended for: UG

Distribution: Elective

Semester: 5th and 7th semester

Approval: 9th Senate

Course Contents

- **Fundamentals of measuring instruments:** Terminology, Units and standards, functional elements of an instrument, input-output configuration, static characteristics and static calibration - least squares calibration curves -static sensitivity - linearity, dynamic characteristics - dynamic response analysis - operational and sinusoidal transfer function - zero-, first-, and second-order instruments, Errors and handling of errors in measurement. [10 lectures]
- **Motion and dimensional measurement:** Relative displacement and velocity: translational and rotational - resistive potentiometers - resistance strain gauge - differential transformers - eddy current noncontacting transducers - ultrasonic transducers - mechanical flyball angular velocity sensor - mechanical revolution counters and timers, relative acceleration, accelerometers - deflection type - null balance type, gyroscopic angular-displacement and velocity sensors, laser rangefinders, ultrasonic rangefinders, radars for automobiles. [12 lectures]
- **Force, torque, and shaft power measurement:** Elastic force transducers - bonded-strain-gauge transducers - differential-transformer transducers - piezoelectric transducers - variable-reluctance, torque measurement on rotating shafts, dynamometers, vibrating-wire force transducers. [3 lectures]
- **Sound and Vibration measurement:** Deadweight gauges and manometers, elastic transducers, vibrating cylinder transducers, high- pressure measurement, low pressure measurement - diaphragm gauges - McLeod gauge - Knudsen gauge - momentum transfer gauges - thermal conductivity gauges - ionisation gauges - dual gauge technique, sound measurement - microphones - pressure response - acoustic intensity - acoustic emission. [3 lectures]
- **Temperature and heat-flux measurement:** Thermal expansion methods - bimetallic thermometers - liquid-in-glass thermometers - pressure thermometers, thermoelectric sensors, electrical resistance sensors - conductive - bulk semiconductor, junction semiconductor sensors, digital thermometers, radiation methods - detectors - automatic null balance - optical - two color - fluoroptic - infrared imaging. [5 lectures]
- **Manipulation, transmission and recording of data:** Bridge circuits, amplifiers, filters, integration and differentiation, cable transmission, fiber- optic data, radio telemetry, Pneumatic transmission, instrument connectivity, potentiometers, digital voltmeters and multimeters, electromechanical servotype XT and XY recorders, data acquisition systems. [4 lectures]
- **Electrical and Electronic Measurements:** Signal generation, Principle of LCR meter, Probe compensation, Spectrum analysis, Instrument Transformers, Measurement of Power and Wattmeters, Measure-

ment of Energy and Industrial Metering, Digital Storage Oscilloscope, Issues of sampling, memory, ADC speed. [6 lectures]

Textbooks:

1. E. O. Doebelin, D. N. Manik, **Measurement systems Application and Design**, 5th Edition, McGraw Hill Book Company, 2007.
2. D. Patranabis, **Principles of Industrial Instrumentation**, 3rd Edition, Tata McGraw Hill Publishing Ltd., 2010.

References:

1. A.K. Sawhney, **A course in electrical and electronic measurements and instrumentation**, 19th Edition, Dhanpat Rai Publications, 2011.
2. A. S. Morris, **Measurement and Instrumentation Principles**, 3rd Edition, Butterworth-Heinemann, 2001.
3. J. P. Holman, **Experimental Methods for Engineers**, McGraw Hill Book Company, 1971.
4. W. C. Dunn, **Fundamentals of Industrial Instrumentation and Process Control**, McGraw Hill, 2005.